

Lab 6, Act 1: Answer Key

Reading the Surface

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (the “which is honest” part of 1a, the explanation in 1b, the judgment part of 2a, and the last part of 2b) get sample answers or discussion points instead. All numbers are computed in R when this file is knitted.

Model: $\text{killed}_i \sim \text{Binomial}(n_i, p_i)$ with $\log \frac{p_i}{1-p_i} = a + bx_i$, $x_i = \log_{10} \text{dose}$. MLEs: $\hat{a} = -60.72$, $\hat{b} = 34.27$. The whole Act is pen and paper; students read values off the printed figures and use a calculator for Part 2.

Part 1: The likelihood surface

1a.

Estimate two intervals for b by eye from the figure (rough values are fine). The **slice** interval: hold a at $\hat{a}$ and find the range of b that stays inside the thick curve. The **profile** interval: the full range of b values the thick curve reaches. Which interval is honest, and why?

Slice: _____ to _____ Profile: _____ to _____

Answer. Slice: about 34.13 to 34.41 (by eye, “about 34.3, basically a single point” is a full-credit answer). Profile: about 28.9 to 40.3 (anything from roughly 28-29 to 40-41 is fine). The profile interval is the honest one.

Why. The slice asks how far b can move with a frozen at $\hat{a}$. Because the valley is so thin and tilted, b can barely move before the fit gets much worse. But a is uncertain too; if b changes, a can change with it and the fit stays good. The profile lets a adjust for every value of b , which is what the extreme top and bottom points of the thick curve represent.

Watch for. Reading the slice as the width of the whole ellipse along some diagonal. Students who say the slice is honest “because it uses the best a ” have missed that $\hat{a}$ is itself uncertain; this is the misconception to target in the group discussion.

1b.

The intercept a is the log-odds of death at what dose (as x , and in mg/L)? Use your answer to explain why the valley is a long, thin diagonal: why does every good fit with a steeper slope also need a more negative intercept?

Answer. a is the log-odds of death at $x = 0$, a dose of $10^0 = 1$ mg/L, far below the tested doses (49-77 mg/L). Every good fit has to pass through roughly 50% mortality near the middle of the data ($x \approx 1.77$). A line that stays near that point but gets steeper must drop further by the time it reaches $x = 0$, so a larger b needs a more negative a .

Why. At the LD50, $a + bx_{50} = 0$, so $a = -bx_{50}$. Along the valley, a change in b is matched by a change in a about 1.77 times as large in the opposite direction, which is the slope of the thick curve on the figure. Because $x = 0$ is so far from the data, a tiny change in b swings a a long way, which is why the valley is so long and thin.

Watch for. Answering “at the lowest dose tested.” Some students say “ a has no biological meaning” and stop; push them to say where $x = 0$ is relative to the data. A good group-discussion prompt: if we measured dose so that $x = 0$ fell in the middle of the data, what would the surface look like? (Part 2 answers this.)

Part 2: Curvature with two parameters

2a.

Compute SE_1 , r , and SE_2 , and the approximate 95% interval $\hat{b} \pm 1.96 SE$ for each (use $\hat{b} = 34.27$). Which do you think better characterizes our uncertainty about b , and why? Relate your answer to Part 1.

Answer. $SE_1 = 1/\sqrt{0.145} = 2.63$; $r = 1.26/\sqrt{58.5 \times 0.145} = 0.433$; $SE_2 = 2.63/\sqrt{1 - 0.433^2} = 2.91$. Intervals: 29.1 to 39.4 using SE_1 ; 28.6 to 40.0 using SE_2 . SE_2 is the better description of uncertainty: its interval is close to the profile interval from 1a (28.9 to 40.3).

Why. SE_1 uses only the curvature along the b axis, with a' held fixed: it is the quadratic version of the slice. SE_2 accounts for the fact that a' can shift to compensate when b moves, as the profile does. The correction factor $1/\sqrt{1 - r^2}$ is always at least 1, so ignoring the correlation can only make us overconfident. (For two parameters, SE_2 is what `sqrt(diag(solve(H)))` returns; with the unrounded Hessian it is 2.91.)

Watch for. Sign of r : H_{ab} is negative, so r is positive (0.433). Getting the sign wrong doesn't change SE_2 because r is squared, but it's worth checking. Students who pick SE_1 “because it's simpler” or “because it's smaller, so more precise” are the ones to talk to: smaller isn't better if it ignores real uncertainty.

2b.

For the original, uncentered fit, $H_{bb} = 185$ and $r = -0.99968$. Compute both standard errors and their intervals again. Compare them with your slice and profile intervals from 1a, and with your answers to 2a. Centering didn't change the fitted curve, so what did it change?

Answer. $SE_1 = 1/\sqrt{185} = 0.0735$, interval 34.13 to 34.41. $SE_2 = 0.0735/\sqrt{1 - 0.99968^2} = 2.91$, interval 28.6 to 40.0. The SE_1 interval matches the slice from 1a; the SE_2 interval is close to the profile, and SE_2 is essentially the same as in 2a.

Centering changed the **shape and tilt of the surface** (correlation -0.9997 before, 0.43 after) and the meaning of the intercept (a' is the log-odds of death at $x = 1.8$, about 63 mg/L, inside the data). It did not change the fitted curve, the maximum likelihood, $\hat{b}$, or the honest uncertainty in b .

Why. SE_1 depends on how the model is written (it changes by a factor of about 40 here), so it can't be a property of what the data say about b . SE_2 and the profile describe the same quantity either way. The inflation factor $1/\sqrt{1 - r^2} \approx 40$ is the numerical version of the near-point slice in 1a.

Watch for. Rounding r to -0.9997 or -1 before squaring: with $r = -0.9997$, SE_2 comes out near 3.0; with $r = -1$ it's undefined. Worth pointing out that when estimates are this correlated, the arithmetic becomes touchy, which is one practical reason to center. Some students will also say centering “made b more precise”; it didn't, it only made SE_1 less misleading.

Discussion point (optional). A thin, tilted valley is also hard for optimizers to walk down, so centering or rescaling predictors often helps fitting as well. This comes back when we cover optimization.

Group discussion and revision

No fixed answer. The most useful disagreements are likely to be in 1b (what a means and why it trades off with b) and in 2a (why the smaller standard error isn't the better one). Look for reflections that connect the slice/profile picture in Part 1 to SE_1/SE_2 in Part 2.